SE Lab 3 — Architecture Selection & Justification

Scenario: Self-Service Coffee Kiosk System

Architecture selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

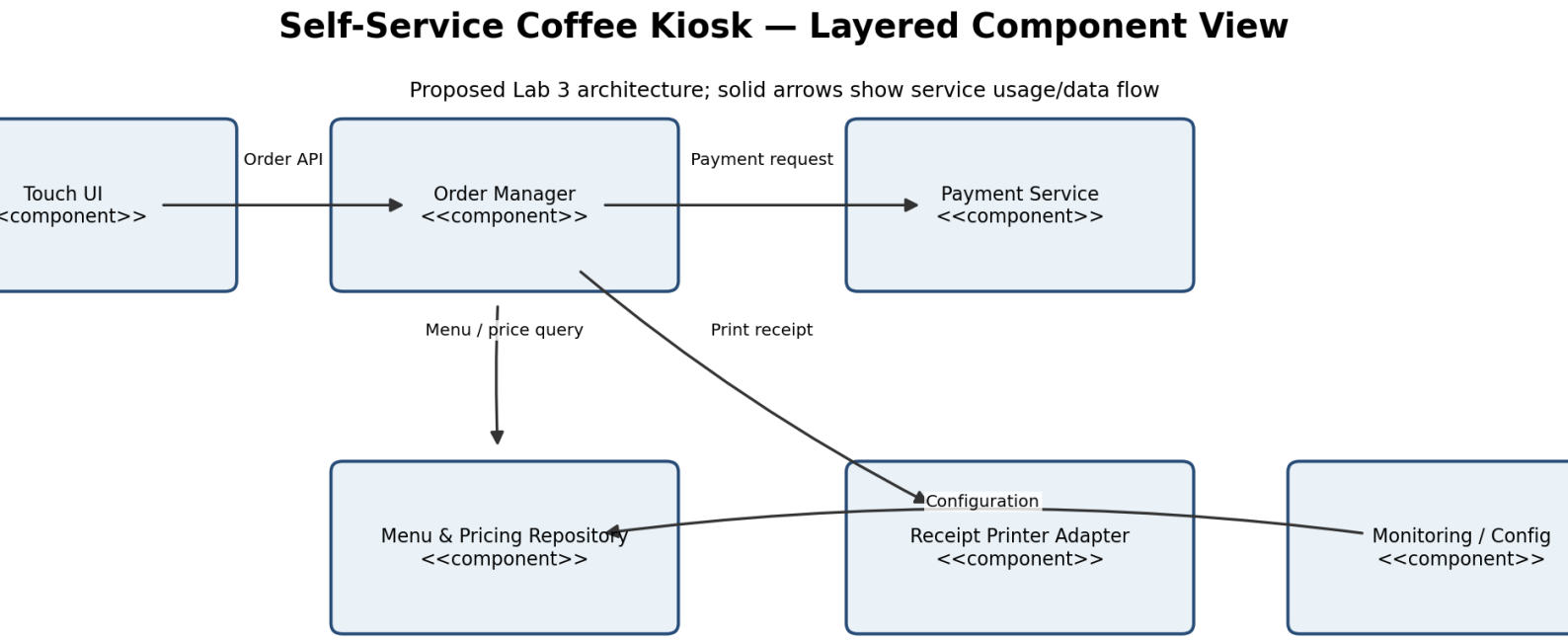
Two reasons. (1) The design separates touch-screen interaction, order coordination, payment, receipt printing and menu/pricing data. This improves readability, testability and maintainability. (2) The kiosk has a small, focused workflow and does not need the operational complexity of independently deployed microservices. Components can still be replaced behind clear interfaces.

Security advantage. Isolating payment behind a dedicated service interface supports using an approved payment terminal/provider without storing raw card details. Input validation and restricted access to configuration help prevent malformed orders and unauthorized price changes.

Performance benefit. Menu and pricing data can be read from a local repository, reducing unnecessary network hops during browsing. The Order Manager validates an order before initiating payment and receipt printing.

Components: Touch UI, Order Manager, Payment Service, Receipt Printer Adapter, Menu & Pricing Repository, Monitoring / Config. The diagram shows labelled data/service flows between components.

Note: If formal UML ball-and-socket notation is explicitly assessed, refine the diagram in a UML editor before submission.



Presentation (Touch UI) → Business (Order Manager, Payment, Printer Adapter) → Data/configuration.

okenize/avoid storing card data; use payment-provider interface; validate inputs and restrict configuration access.